

Deep Learning Assessment of Metastatic Relapse Risk from Digitized Breast Cancer Histological Slides.

Corresponding Author: Dr Ingrid Garberis

This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

In this manuscript Garberis et al. describes a deep learning-based method (RlapsRisk BC) for breast cancer risk stratification to assess 5-year metastasis-free survival in ER+HER2- early breast cancer. The manuscript has already been reviewed by three reviewers at Nature Medicine, and reviewer suggestions have been incorporated by the authors. Two major points were addressed in response to comments by reviewer #1: the authors stained the full training and part of the validation cohort with the standard HE staining (instead of HES staining), making the results more widely applicable. They also incorporated a large independent validation cohort from a different centre than the training cohort. Those two additions led to more generalisability between institutions and wide applicability of the study results, and made the study considerably stronger. In addition, the authors addressed a variety of other reviewer's suggestions, which made the study stronger. However, I have a few outstanding comments:

- RlapsRiskBC is mentioned without any reference in the first paragraph of the results section. Please explain here what RlapsRiskBC is.
- In the introduction the online tool Predict Breast Cancer is mentioned, how does RlapsRiskBC compare to Predict in the datasets used? This would be a relevant addition to the manuscript.
- Table 1C: 'intermediate clinical risk group'. Please clarify here which criteria are used for the intermediate risk definition.
- The survival curves in Figure 3 and Figure 4 should be enlarged.
- There are now many display items (5 figures, many tables), tables are also divided in A, B, C. I would suggest that the authors focus on the most important display items for the main manuscript and put the other tables in the supplements. In addition, there is now a table 3A, table 3B and table 3.

(Remarks on code availability)

Reviewer #5

(Remarks to the Author)

In their paper, the authors developed an AI-based tool, RlapsRisk® BC, to predict 5-year metastasis-free survival (MFS) in ER+/HER2- early breast cancer using digitized tumor slides of H&E and HES. Their main goals, supported by their analysis, were to demonstrate that their RlapsRisk® BC model based on digital slides alone, and their model based on combined pathology slides and clinico-pathological factors, predict risk stratification better than using clinico-pathological factors alone. The authors used slides from two independent cohorts, with a total of approximately 2500 patients. While their study shows that their model extracts prognostic information from pathology slides that is additive to clinico-pathology features, I have several concerns regarding the originality and implications of their study.

1) Originality and contributions over previous studies

While the authors claim novelty in their approach, many prior studies have demonstrated the prognostic value of deep

learning applied to H&E slides in breast cancer and other cancers. The authors' claim that H&E slides contain prognostic information beyond clinico-pathology is well established in the literature, and the discussion does not sufficiently acknowledge this. For example:

- "Contrary to other digital pathology approaches predicting morphological features, such as histological grade or KI67 index (25),(26),(27),(26) or based on tumor micro-environment features (20) that are ultimately indirectly linked to prognosis, we trained our Deep Learning model to directly predict the 5-year MFS. Not only did we not require local annotations to train the algorithm, which takes as input the entire WSI, but also our prediction task was directly addressing our clinical question, the prediction of 5-years MFS". (with 26 shown twice for some reason). The authors state that their approach is novel because it directly predicts 5-year MFS without features and without requiring local annotations. However, However, papers 25-27 are from 2018, and many studies since 2018 have already predicted recurrence or other prognostic measures from H&E images directly without requiring annotations. paper 20 is from 2024 and is indeed novel - but because of its interpretive features and not despite.

- Examples of recent relevant studies:

-

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<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10640636/>

Nov 2023

Predicting early breast cancer recurrence from histopathological images in the Carolina Breast Cancer Study.

<https://link.springer.com/article/10.1007/s10549-024-07303-z#Sec7>

April 2024

Clinical evaluation of deep learning-based risk profiling in breast cancer histopathology and comparison to an established multigene assay.

<https://www.nature.com/articles/s41698-023-00472-y>

Nov 2023

AI-enabled routine H&E image based prognostic marker for early-stage luminal breast cancer.

<https://link.springer.com/article/10.1186/s13058-024-01879-6>

Aug 2024

Validation of an AI-based solution for breast cancer risk stratification using routine digital histopathology images

<https://www.nature.com/articles/s41523-023-00545-y#Sec8>

17 May 2023

Computational pathology improves risk stratification of a multi-gene assay for early stage ER+ breast cancer

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0233678>

2020

Deep learning-based survival prediction for multiple cancer types using histopathology images

<https://link.springer.com/article/10.1007/s10549-019-05281-1>

2019

Breast cancer outcome prediction with tumour tissue images and machine learning

<https://www.nature.com/articles/s41523-023-00530-5#MOESM1>

April 2023

Integration of clinical features and deep learning on pathology for the prediction of breast cancer recurrence assays and risk of recurrence

o Studies in other cancers, such as prostate cancer, have also demonstrated the same methodological and prognostic principles (e.g., Nature Digital Medicine, June 2022).

o Without a comprehensive literature review, the authors fail to contextualize their contribution within the field or clarify how their approach advances the state of the art.

2) Implications and Utility

The authors suggest that the presented system could provide better patient stratification than current methods, such as gene expression profiling or clinical features. For example, they state:

"The discrepant results of different tests and their restrictive indications limit their use, while their cost is currently not covered by the health care system in many countries."

"Our system could support decision-making, replace molecular signatures or, at least, work as a pre-screening test."

However, the paper does not provide evidence for these claims:

- There is no comparative analysis showing improvement over existing molecular tests (e.g., Oncotype DX).
- The authors do not present cost-effectiveness data to support their claim, which is especially relevant in countries without genomic test coverage.
- There is no discussion on the methodology of pre-screening or threshold setting for endorsing genomic tests. One would need to define a threshold below or above which patients would be excluded/endorsed for the expansive genomic tests.
- The authors do not discuss the limitations of AI models for clinical decision-making, including interpretability and the risks of domain shift.
- Without randomized treatment studies, a prognostic model has limited utility, as confounding factors may prevent actionable outcomes.

- Finally, the authors refrain from publishing their data, model, or code, which further limits the implications of their work.

3) Methodology

The methodology relies heavily on existing techniques:

- Feature extraction is based on SSL (43), and the downstream task uses the well-established MIL strategy (46) and smoothed c-index loss (47). These approaches are not novel.
- The authors do not clarify why they cannot publish their model or code if it is based on prior methods.
- The SSL feature extractor was changed during the review process: "We thank the reviewers for this valuable feedback which led us to retrain new feature extractors using new state-of-the-art methods: IBOT Vit Base."

This raises the question: What exactly is RlapsRisk® BC? Is it an established method?

A major concern is the lack of clarity about how the combined model using both image-based and clinico-pathological features was created and trained. Were these features combined during training? If it was evaluated together in the multivariate analysis on the test set, the analysis may not be sound.

4) Soundness of Analysis and Reproducibility

The authors' analysis appears valid and provides evidence for their main claim that H&E/HES images add prognostic information over clinico-pathological variables using the proposed model. Specifically, regarding reviewer 2's comment on the 3-fold cross-validation, the newly proposed method of using 10,000 permutations to compute the p-value is valid.

However, several issues remain:

- The fact that the code, models, and data are not published significantly limits the soundness and reproducibility of the study.
- The test on the CANTO dataset involves a mix of two populations (different slides and cohorts), rather than analyzing each cohort separately, which could introduce confounding factors.
- Details on the analysis are incomplete. For example:
 - o There is insufficient information on the training process. The authors mention: "RlapsRisk BC scores are derived from the validation splits of the repeated cross-validation training procedure," but there is no detailed discussion of cross-validation, folds, or splits across the paper.
 - o The authors refer readers to another paper to understand the model and loss function. However, these details are critical for understanding and replicating the study, and their absence in the manuscript is a notable shortcoming.

5) Writing Quality

Although this is not a major concern, I found the paper to be not fluent, with missing details, difficult to read, and occasionally containing typos. Some examples include:

- Unclear sentence structure:
 - o "This higher risk of the patients with chemotherapy being captured by the clinical variables in our multivariable analyses, we excluded the variable representing the adjuvant chemotherapy treatment regimen from all prognostic models to avoid redundancy of information that could lead to multicollinearity issues."

This sentence structure is convoluted and unclear.

- o "We considered this score as our baseline reference that we compared to a score based on a Cox Model adjusted for the clinical factors and RlapsRisk BC score to assess the additional prognostic value of the RlapsRisk BC score to the standard clinical factors."

This sentence is unnecessarily long and difficult to follow.

- Misleading terminology:
 - o If the data consists of whole slide images, why is the cohort referred to as "GrandTMA"? The name is misleading and could cause confusion for readers expecting tissue microarray (TMA) data.
- There are many similar instances where clarity could be improved, and addressing these issues would significantly enhance the readability and accessibility of the paper.

(Remarks on code availability)

Reviewer #6

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

The authors addressed all my comments.

(Remarks on code availability)

Reviewer #5

(Remarks to the Author)

I appreciate the authors' detailed responses and the clarifications in the Discussion section.

However, I still have concerns about the methodological details, which are critical for reproducibility—especially since the authors are unable to publish their code. Below are specific points that need further elaboration: In a smoothed c-index approach, the computational complexity is $O(N^2)$ in the number of slides N . When each slide has many tiles, memory usage can become prohibitive. There is also the practical requirement of ensuring that each minibatch contains enough events for stable gradient updates. If events occur only every 20–30 patients, what was the average number of events per minibatch in practice and what was the minibatch size? Clarifying how the minibatch size and tile sampling per slide were chosen and how memory constraints were mitigated would help readers reproduce the method. How exactly was the end-to-end pipeline set up? Were there any techniques such as pairwise sub-sampling for the c-index, or any special strategies to handle large risk sets? The paper should specify any memory-optimization tricks. The authors mention using a smoothed c-index loss, but it would be helpful to understand the reasons for preferring this over a Cox partial-likelihood loss. Overall, the study's topic is important, and the authors' work could be valuable if these clarifications are provided. Without these methodological details, reproducing and extending the findings would be challenging.

(Remarks on code availability)

See above

Reviewer #6

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

The methodological code was not available

Version 2:

Reviewer comments:

Reviewer #5

(Remarks to the Author)

The new manuscript addresses all the concerns

(Remarks on code availability)

Reviewer #6

(Remarks to the Author)

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(Remarks on code availability)

No code available

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Response to Referees

Reviewers Comments and answers in blue

We appreciate the feedback provided by the reviewers and the opportunity to clarify key aspects of our study. Below, we address the specific concerns raised:

Comparison with Existing Tools (Reviewer #4):

- We have expanded our discussion on how the combined model compares with Predict Breast, a clinically established risk stratification tool. In our revised manuscript, we provide additional performance metrics and a more detailed analysis highlighting the added value of integrating an histopathological score on top of clinical features. This information is now included in the Results section and reads as follows: **“We then perform a comparison with the Predict Breast (PB) model, which is a prognostic tool that estimates survival outcomes based on clinical and treatment-related factors (53). The RlapsRisk combined model outperforms all PB-based survival models, achieving the highest C-index (0.81 [0.78–0.86]) (Supplementary Table 5). In contrast, PB models show lower predictive performance, with C-index values ranging from 0.71 to 0.76”**

Cross-Validation Methods (Reviewer #5):

- We have clarified our cross-validation approach, ensuring transparency in how we partitioned the dataset and validated model performance. Specifically, we now describe the stratified k-fold cross-validation procedure used to prevent data leakage and ensure robustness. Further details on the validation process, including steps taken to mitigate overfitting, are now outlined in the Methods Section and reads as follows : **“The RlapsRisk BC risk score was optimized using the discovery cohort GrandTMA (HES WSIs) with a stratified fivefold cross-validation approach, repeated three times. Due to the limited number of events, stratification was performed based on event occurrence to ensure a minimum number of events in each fold.”**

REVIEWER COMMENTS

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In this manuscript Garberis et al. describes a deep learning-based method (RlapsRisk BC) for breast cancer risk stratification to assess 5-year metastasis-free survival in ER+HER2- early breast cancer. The manuscript has already been reviewed by three reviewers at Nature Medicine, and reviewer suggestions have been incorporated by the authors. Two major points were addressed in response to comments by reviewer #1: the authors stained the full training and part of the validation cohort with the standard HE staining (instead of HES staining),

making the results more widely applicable. They also incorporated a large independent validation cohort from a different centre than the training cohort. Those two additions led to more generalisability between institutions and wide applicability of the study results, and made the study considerably stronger. In addition, the authors addressed a variety of other reviewer's suggestions, which made the study stronger. However, I have a few outstanding comments:

- RlapsRiskBC is mentioned without any reference in the first paragraph of the results section. Please explain here what RlapsRiskBC is.

The first sentence now reads as 'The primary objective of this study was to evaluate the additional 5-years MFS prognostic value of RlapsRisk BC score, **an AI-derived prognostic score based exclusively on WSI**, relative to that of the current clinico-pathological criteria [...]'

- In the introduction the online tool Predict Breast Cancer is mentioned, how does RlapsRiskBC compare to Predict in the datasets used? This would be a relevant addition to the manuscript.

We have expanded our discussion on how the combined model compares with Predict Breast, a clinically established risk stratification tool. In our revised manuscript, we provide additional performance metrics and a more detailed analysis highlighting the added value of integrating an histopathological score on top of clinical features. This information is now included in the Results section and reads as follows: **"We then perform a comparison with the Predict Breast (PB) model, which is a prognostic tool that estimates survival outcomes based on clinical and treatment-related factors (SK Down et al, 2014). The RlapsRisk combined model outperforms all PB-based survival models, achieving the highest C-index (0.82 [0.78–0.86])"**

- Table 1C: 'intermediate clinical risk group'. Please clarify here which criteria are used for the intermediate risk definition.

The intermediate risk group is defined in supp Table 2 and this is mentioned in the manuscript 'This comparison was assessed on the entire population of the validation cohort and in different subgroups of clinical interest (histological grade 2, **intermediate clinical risk of relapse as defined in the French guidelines displayed in supplementary table 2**, pre-versus post-menopausal status, with or without lymph node invasion, treated with or without adjuvant chemotherapy).'

- The survival curves in Figure 3 and Figure 4 should be enlarged.
We have reworked the figures and made them easier to read.

- There are now many display items (5 figures, many tables), tables are also divided in A, B, C. I would suggest that the authors focus on the most important display items for the main manuscript and put the other tables in the supplements. In addition, there is now a table 3A, table 3B and table 3.

We have streamlined the main corpus by removing numerous figures and tables, retaining only those essential for a comprehensive understanding.
There are now 4 figures and 4 tables.

Reviewer #5 (Remarks to the Author):

In their paper, the authors developed an AI-based tool, RlapsRisk® BC, to predict 5-year metastasis-free survival (MFS) in ER+/HER2- early breast cancer using digitized tumor slides of H&E and HES. Their main goals, supported by their analysis, were to demonstrate that their RlapsRisk® BC model based on digital slides alone, and their model based on combined pathology slides and clinico-pathological factors, predict risk stratification better than using clinico-pathological factors alone. The authors used slides from two independent cohorts, with a total of approximately 2500 patients. While their study shows that their model extracts prognostic information from pathology slides that is additive to clinico-pathology features, I have several concerns regarding the originality and implications of their study.

1) Originality and contributions over previous studies

While the authors claim novelty in their approach, many prior studies have demonstrated the prognostic value of deep learning applied to H&E slides in breast cancer and other cancers. The authors' claim that H&E slides contain prognostic information beyond clinico-pathology is well established in the literature, and the discussion does not sufficiently acknowledge this. For example:

- "Contrary to other digital pathology approaches predicting morphological features, such as histological grade or KI67 index (25),(26),(27),(26) or based on tumor micro-environment features (20) that are ultimately indirectly linked to prognosis, we trained our Deep Learning model to directly predict the 5-year MFS. Not only did we not require local annotations to train the algorithm, which takes as input the entire WSI, but also our prediction task was directly addressing our clinical question, the prediction of 5-years MFS". (with 26 shown twice for some reason). The authors state that their approach is novel because it directly predicts 5-year MFS without features and without requiring local annotations. However, papers 25-27 are from 2018, and many studies since 2018 have already predicted recurrence or other prognostic measures from H&E images directly without requiring annotations. paper 20 is from 2024 and is indeed novel - but because of its interpretive features and not despite.

- Examples of recent relevant studies:

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<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10640636/>

Nov 2023

Predicting early breast cancer recurrence from histopathological images in the Carolina Breast Cancer Study.

Integration of clinical features and deep learning on pathology for the prediction of breast cancer recurrence assays and risk of recurrence

- o Studies in other cancers, such as prostate cancer, have also demonstrated the same methodological and prognostic principles (e.g., Nature Digital Medicine, June 2022).

o Without a comprehensive literature review, the authors fail to contextualize their contribution within the field or clarify how their approach advances the state of the art. (à finir de modifier)

We thank the reviewer for this comment. The lack of novelty of our work suggested here (as well as the date of some of the articles in the bibliography, which we have updated) is partly due to the fact that our first submission to Nature Medicine dates from February 2023. This revision took considerable time and a great workload from our team to enrich our approach and to address the reviewers' recommendations, which we resubmitted in April 2024. In addition, our research is framed within a prolific field, targeting a prevalent disease that represents a true global issue, which multiplies the number of working groups and efforts looking for a potential improvement.

This being said, we have now considerably updated this part of the discussion accounting for the references provided by the reviewer. We now fully acknowledge that studies in other cancers, such as prostate cancer, have also demonstrated the same methodological and prognostic principles. This part of the discussion now reads as follows : "Our approach, which directly predicts a survival endpoint from H&E slides, has already been successfully applied in prostate cancer. Notably, AI models using digital histopathology analysis have improved prognostic stratification and therapy personalization in prostate cancer, outperforming standard classification tools (REF). These findings support the relevance of applying a similar approach in breast cancer, where integrating AI into digital pathology could further refine risk assessment."

2) Implications and Utility

The authors suggest that the presented system could provide better patient stratification than current methods, such as gene expression profiling or clinical features. For example, they state:

"The discrepant results of different tests and their restrictive indications limit their use, while their cost is currently not covered by the health care system in many countries."

"Our system could support decision-making, replace molecular signatures or, at least, work as a pre-screening test."

However, the paper does not provide evidence for these claims:

- There is no comparative analysis showing improvement over existing molecular tests (e.g., Oncotype DX).
- The authors do not present cost-effectiveness data to support their claim, which is especially relevant in countries without genomic test coverage.
- There is no discussion on the methodology of pre-screening or threshold setting for endorsing genomic tests. One would need to define a threshold below or above which patients would be excluded/endorsed for the expansive genomic tests.

We acknowledge that this part of the discussion was not fully supported by the results presented. To provide a perspective on the potential pre-screening application of our test without overstating its clinical utility— which has yet to be established— we have substantially revised this section. We have softened our critique of genomic tools, refined our wording, and incorporated a new reference on digital pathology for pre-screening (Boehm, Kevin M., et al. 'Multimodal histopathologic models stratify hormone receptor-positive early breast cancer.' BioRxiv (2024): 2024-02). The revised paragraph now reads as follows:

"Multimodal histopathologic models have been investigated as a means to stratify hormone receptor-positive early breast cancer and could serve as pre-screening tools to identify patients who may or may not require more expensive genomic testing (Boehm et al 2024). While our study does not provide a direct comparison with genomic assays such as Oncotype DX, our results suggest that a WSI-based approach could serve as a complementary tool in clinical decision-making. Head-to-head comparison of RlapsRisk BC and molecular signatures in a cohort similar to TRANSATAC would provide further insights into its potential clinical utility, including defining optimal thresholds for genomic test endorsement (Sestak et al. 2018). In addition, assessment of the impact of a decision-making strategy based on RlapsRisk BC would require a randomized clinical trial to eliminate potential biases that may arise from observational studies."

- The authors do not discuss the limitations of AI models for clinical decision-making, including interpretability and the risks of domain shift.

Domain shift is indeed a significant concern in digital pathology, and we now explicitly address it in the Discussion as follows: "To use RlapsRisk BC in clinical practice, the calibration protocol requires 30 BC slides, which may pose challenges, particularly in smaller pathology laboratories. While calibration is a standard procedure for many medical devices, it could impair the clinical deployment of such AI tools. We have added the following paragraph in the discussion section

"Furthermore, variations in staining protocols, scanner settings, and other technical factors between healthcare centers could introduce domain shifts, potentially affecting the model's reliability. Here prognosis results were robust with respect to the staining protocols (H&E and HES) since concordant prognostic results were found in two centers using two different staining protocols. However, the diversity of worldwide staining protocols is larger than the two protocols tested here, stressing the importance of analytical validation."

Interpretability is already addressed in another paragraph and we now have considerably revised the paragraph about interpretability that now reads as follows

"The interpretability analysis of the pathological features associated with an increased or decreased risk of relapse, as predicted by RlapsRisk BC, supports the validity of our model. The model's reliance on well-established histological prognostic factors—such as nuclear pleomorphism, tumor architecture, and mitotic activity—demonstrates its alignment with known pathology-driven risk assessments."

- Without randomized treatment studies, a prognostic model has limited utility, as confounding factors may prevent actionable outcomes.

We add the following sentence to acknowledge the limitation of our study **"In addition, assessment of the impact of a decision-making strategy based on RlapsRisk BC would require a randomized clinical trial to eliminate potential biases that may arise from observational studies."** to acknowledge this limitation."

- Finally, the authors refrain from publishing their data, model, or code, which further limits the implications of their work.

The scripts used to run the training and inference of the featured model are not public as they rely heavily on the intellectual property of Owkin, Inc. and on proprietary libraries. Those libraries include a tool to preprocess and tile raw whole-slide images; and a library of weakly supervised machine learning models. Nonetheless we provided the scripts used to generate the results presented in our manuscript.

3) Methodology

The methodology relies heavily on existing techniques:

- Feature extraction is based on SSL (43), and the downstream task uses the well-established MIL strategy (46) and smoothed c-index loss (47). These approaches are not novel.

Our study focuses on the extensive evaluation of RlapsRisk BC in a clinical setting rather than developing novel ML techniques. While we build on established methods like SSL (43), MIL (46), and smoothed c-index loss (47), our key contribution lies in demonstrating the real-world applicability of AI-driven prognostic modeling through rigorous validation.

- The authors do not clarify why they cannot publish their model or code if it is based on prior methods.

As already mentioned in a previous review and again here at the end of “2) Implications and Utility”, we are constrained from publishing the model or code due to intellectual property issues (Owkin, Inc. being a private company). We understand the importance of transparency and replicability in scientific research. While we are unable to release the code due to proprietary reasons, we are committed to upholding the highest standards of integrity and rigor in our methodologies. We recognize the significance of replicability in advancing the field and will continue to explore ways to contribute to this goal within the constraints of our business model. It's essential to address the concerns raised, and we remain dedicated to mitigating any issues identified to ensure the credibility and trustworthiness of our work.

- The SSL feature extractor was changed during the review process: "We thank the reviewers for this valuable feedback which led us to retrain new feature extractors using new state-of-the-art methods: IBOT Vit Base."

This raises the question: What exactly is RlapsRisk® BC? Is it an established method?

RlapsRisk BC is an evolving AI-based prognostic tool, and like many IVD SaMD (In Vitro Diagnostic Software as a Medical Device) solutions, it undergoes continuous improvement. The version presented in this study reflects the most up-to-date model, incorporating advancements such as IBOT ViT Base for SSL feature extraction.

A major concern is the lack of clarity about how the combined model using both image-based and clinico-pathological features was created and trained. Were these features combined

during training? If it was evaluated together in the multivariate analysis on the test set, the analysis may not be sound.

We acknowledge that the presentation of the Combined score was not clear enough. We added a few sentences to precise how the Combined score was obtained, it now reads as follows in the *Model development and datasets* section of the Results: **“The coefficients of this Cox model, which provides the Combined Score, were also trained on the discovery dataset.”** We confirm that the Combined score is obtained by using only models fitted on the discovery dataset.

To avoid any misunderstanding, we also added a precision “RlapsRisk BC score and Prognosis” section of the Results on the multivariable analyses, where indeed the HRs are computed using the external validation cohort : **“RlapsRisk BC score was an independent prognostic factor of MFS (Table 1A and Supplementary Table 3A) when integrated into a multivariable Cox model including histological grade, age, lymph node invasion, tumor size, and Ki67 expression directly fitted on cohorts”.**

4) Soundness of Analysis and Reproducibility

The authors' analysis appears valid and provides evidence for their main claim that H&E/HES images add prognostic information over clinico-pathological variables using the proposed model. Specifically, regarding reviewer 2's comment on the 3-fold cross-validation, the newly proposed method of using 10,000 permutations to compute the p-value is valid. However, several issues remain:

- The fact that the code, models, and data are not published significantly limits the soundness and reproducibility of the study.
- The test on the CANTO dataset involves a mix of two populations (different slides and cohorts), rather than analyzing each cohort separately, which could introduce confounding factors.

As described in the supplementary material, separate analyses yield results consistent with those obtained from the entire cohort. These findings are presented in the supplementary section titled 'Results on the CANTO Cohort per Staining Method,' specifically in Tables 4A and 4B, which detail the multivariable analyses stratified by population.

- Details on the analysis are incomplete. For example:

There is insufficient information on the training process. The authors mention: "RlapsRisk BC scores are derived from the validation splits of the repeated cross-validation training procedure," but there is no detailed discussion of cross-validation, folds, or splits across the paper.

We acknowledge that the details on the training procedure were missing, it now reads as follows : **“The RlapsRisk BC risk score was optimized using the discovery cohort GrandTMA (HES WSIs) with a stratified fivefold cross-validation approach, repeated**

three times. Given the limited number of events, stratification was performed based on event occurrence to ensure a minimum number of events in each fold.”

o The authors refer readers to another paper to understand the model and loss function. However, these details are critical for understanding and replicating the study, and their absence in the manuscript is a notable shortcoming.

A more detailed description of the functioning DeepMIL has been added to the section called “Risk Prediction using Multiple Instance learning” of the methods, it now reads as follows : **“DeepMIL is a deep multiple instance learning (MIL) framework designed for weakly supervised classification. It aggregates instance-level (the WSI’s tiles in our case) features using an attention-based pooling mechanism, which assigns learned importance weights to each instance within a bag. This allows the model to focus on the most relevant instances for prediction. The weighted instance representations are then combined into a single bag-level feature vector, which is processed by a fully connected layer to generate the final classification output. This approach enhances interpretability while effectively handling variability within each bag.”**

We also added a description of the functioning of the loss function in the same paragraph, it now reads as follows: **“We used a smooth differentiable loss function based on a sigmoid approximation of the standard C-index indicator function. This smoothing introduces a parameter σ that controls the transition sharpness, enabling gradient-based optimization. The loss function is integrated into a gradient boosting framework, iteratively minimizing the smoothed empirical risk to improve model discrimination. This approach ensures that the learned biomarker combinations are directly aligned with survival prediction performance.”**

5) Writing Quality

Although this is not a major concern, I found the paper to be not fluent, with missing details, difficult to read, and occasionally containing typos. Some examples include:

- Unclear sentence structure:

o "This higher risk of the patients with chemotherapy being captured by the clinical variables in our multivariable analyses, we excluded the variable representing the adjuvant chemotherapy treatment regimen from all prognostic models to avoid redundancy of information that could lead to multicollinearity issues."

This sentence structure is convoluted and unclear.

It has been modified and now reads as follows **“We excluded chemotherapy treatment from the clinical score because it was prescribed to higher-risk patients based on clinical best practices, introducing prescription bias. Since this risk was already captured through clinical variables, including chemotherapy would have led to redundancy and multicollinearity in the prognostic models.”**

o "We considered this score as our baseline reference that we compared to a score based on a Cox Model adjusted for the clinical factors and RlapsRisk BC score to assess the additional prognostic value of the RlapsRisk BC score to the standard clinical factors."

It has been modified and now reads as follows: **“We used this score as the baseline reference and compared it to a score derived from a Cox model adjusted for clinical factors and the RlapsRisk BC score (referred to as the Combined Score).”**

This sentence is unnecessarily long and difficult to follow.

- Misleading terminology:

- o If the data consists of whole slide images, why is the cohort referred to as "GrandTMA"? The name is misleading and could cause confusion for readers expecting tissue microarray (TMA) data.

The training cohort chosen for this study originates from an institutional project at Gustave Roussy with the aim of creating a large tissue microarray that could allow the discovery and/or validation of biomarkers in a substantial subset of breast cancer samples. We found that this population and the available material (clinical data, histopathological glass slides stored in the pathology department) were perfectly suited to our purpose. Therefore, the name refers to the primary objective of this database and we do not think that we could rename this collection just because we are now using it for a different application than originally designed.

There are many similar instances where clarity could be improved, and addressing these issues would significantly enhance the readability and accessibility of the paper.

The text has now been considerably improved in several places which are indicated in bold face in the revised manuscript.

Reviewer #6 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

We appreciate the feedback provided by you and the reviewers and the opportunity to clarify key aspects of our study.

Response to Referees

Reviewers Comments and answers in blue

Comparison with Existing Tools (Reviewer #4):

- We have expanded our discussion on how the combined model compares with Predict Breast, a clinically established risk stratification tool. In our revised manuscript, we provide additional performance metrics and a more detailed analysis highlighting the added value of integrating an histopathological score on top of clinical features. This information is now included in the Results section and reads as follows: **“We then perform a comparison with the Predict Breast (PB) model, which is a prognostic tool that estimates survival outcomes based on clinical and treatment-related factors (53). The RlapsRisk combined model outperforms all PB-based survival models, achieving the highest C-index (0.81 [0.78–0.86]) (Supplementary Table 5). In contrast, PB models show lower predictive performance, with C-index values ranging from 0.71 to 0.76”**

Cross-Validation Methods (Reviewer #5):

- We have clarified our cross-validation approach, ensuring transparency in how we partitioned the dataset and validated model performance. Specifically, we now describe the stratified k-fold cross-validation procedure used to prevent data leakage and ensure robustness. Further details on the validation process, including steps taken to mitigate overfitting, are now outlined in the Methods Section and reads as follows : **“The RlapsRisk BC risk score was optimized using the discovery cohort GrandTMA (HES WSIs) with a stratified fivefold cross-validation approach, repeated three times. Due to the limited number of events, stratification was performed based on event occurrence to ensure a minimum number of events in each fold.”**

REVIEWER COMMENTS

Reviewer #4 (Remarks to the Author):

In this manuscript Garberis et al. describes a deep learning-based method (RlapsRisk BC) for breast cancer risk stratification to assess 5-year metastasis-free survival in ER+HER2- early breast cancer. The manuscript has already been reviewed by three reviewers at Nature Medicine, and reviewer suggestions have been incorporated by the authors. Two major points were addressed in response to comments by reviewer #1: the authors stained the full training and part of the validation cohort with the standard HE staining (instead of HES staining), making the results more widely applicable. They also incorporated a large independent validation cohort from a different centre than the training cohort. Those two additions led to more generalisability between institutions and wide applicability of the study results, and made

the study considerably stronger. In addition, the authors addressed a variety of other reviewer's suggestions, which made the study stronger. However, I have a few outstanding comments:

- RlapsRiskBC is mentioned without any reference in the first paragraph of the results section. Please explain here what RlapsRiskBC is.

The first sentence now reads as 'The primary objective of this study was to evaluate the additional 5-years MFS prognostic value of RlapsRisk BC score, **an AI-derived prognostic score based exclusively on WSI**, relative to that of the current clinico-pathological criteria [...]'

- In the introduction the online tool Predict Breast Cancer is mentioned, how does RlapsRiskBC compare to Predict in the datasets used? This would be a relevant addition to the manuscript.

We have expanded our discussion on how the combined model compares with Predict Breast, a clinically established risk stratification tool. In our revised manuscript, we provide additional performance metrics and a more detailed analysis highlighting the added value of integrating an histopathological score on top of clinical features. This information is now included in the Results section and reads as follows: **"We then perform a comparison with the Predict Breast (PB) model, which is a prognostic tool that estimates survival outcomes based on clinical and treatment-related factors (SK Down et al, 2014). The RlapsRisk combined model outperforms all PB-based survival models, achieving the highest C-index (0.82 [0.78–0.86])"**

- Table 1C: 'intermediate clinical risk group'. Please clarify here which criteria are used for the intermediate risk definition.

The intermediate risk group is defined in supp Table 2 and this is mentioned in the manuscript 'This comparison was assessed on the entire population of the validation cohort and in different subgroups of clinical interest (histological grade 2, **intermediate clinical risk of relapse as defined in the French guidelines displayed in supplementary table 2**, pre-versus post-menopausal status, with or without lymph node invasion, treated with or without adjuvant chemotherapy).'

- The survival curves in Figure 3 and Figure 4 should be enlarged. We have reworked the figures and made them easier to read.

- There are now many display items (5 figures, many tables), tables are also divided in A, B, C. I would suggest that the authors focus on the most important display items for the main manuscript and put the other tables in the supplements. In addition, there is now a table 3A, table 3B and table 3.

We have streamlined the main corpus by removing numerous figures and tables, retaining only those essential for a comprehensive understanding.

There are now 4 figures and 4 tables.

Reviewer #5 (Remarks to the Author):

In their paper, the authors developed an AI-based tool, RlapsRisk® BC, to predict 5-year metastasis-free survival (MFS) in ER+/HER2- early breast cancer using digitized tumor slides of H&E and HES. Their main goals, supported by their analysis, were to demonstrate that their RlapsRisk® BC model based on digital slides alone, and their model based on combined pathology slides and clinico-pathological factors, predict risk stratification better than using clinico-pathological factors alone. The authors used slides from two independent cohorts, with a total of approximately 2500 patients. While their study shows that their model extracts prognostic information from pathology slides that is additive to clinico-pathology features, I have several concerns regarding the originality and implications of their study.

1) Originality and contributions over previous studies

While the authors claim novelty in their approach, many prior studies have demonstrated the prognostic value of deep learning applied to H&E slides in breast cancer and other cancers. The authors' claim that H&E slides contain prognostic information beyond clinico-pathology is well established in the literature, and the discussion does not sufficiently acknowledge this. For example:

- "Contrary to other digital pathology approaches predicting morphological features, such as histological grade or KI67 index (25),(26),(27),(26) or based on tumor micro-environment features (20) that are ultimately indirectly linked to prognosis, we trained our Deep Learning model to directly predict the 5-year MFS. Not only did we not require local annotations to train the algorithm, which takes as input the entire WSI, but also our prediction task was directly addressing our clinical question, the prediction of 5-years MFS". (with 26 shown twice for some reason). The authors state that their approach is novel because it directly predicts 5-year MFS without features and without requiring local annotations. However, papers 25-27 are from 2018, and many studies since 2018 have already predicted recurrence or other prognostic measures from H&E images directly without requiring annotations. paper 20 is from 2024 and is indeed novel - but because of its interpretive features and not despite.

- Examples of recent relevant studies:

-

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10640636/>

Nov 2023

Predicting early breast cancer recurrence from histopathological images in the Carolina Breast Cancer Study.

Integration of clinical features and deep learning on pathology for the prediction of breast cancer recurrence assays and risk of recurrence

- o Studies in other cancers, such as prostate cancer, have also demonstrated the same methodological and prognostic principles (e.g., Nature Digital Medicine, June 2022).

- o Without a comprehensive literature review, the authors fail to contextualize their contribution within the field or clarify how their approach advances the state of the art. (à finir de modifier)

We thank the reviewer for this comment. The lack of novelty of our work suggested here (as well as the date of some of the articles in the bibliography, which we have updated) is partly due to the fact that our first submission to Nature Medicine dates from February 2023. This revision took considerable time and a great workload from our team to enrich our approach and to address the reviewers' recommendations, which we resubmitted in April 2024. In addition, our research is framed within a prolific field, targeting a prevalent disease that represents a true global issue, which multiplies the number of working groups and efforts looking for a potential improvement.

This being said, we have now considerably updated this part of the discussion accounting for the references provided by the reviewer. We now fully acknowledge that studies in other cancers, such as prostate cancer, have also demonstrated the same methodological and prognostic principles. This part of the discussion now reads as follows : "Our approach, which directly predicts a survival endpoint from H&E slides, has already been successfully applied in prostate cancer. Notably, AI models using digital histopathology analysis have improved prognostic stratification and therapy personalization in prostate cancer, outperforming standard classification tools (REF). These findings support the relevance of applying a similar approach in breast cancer, where integrating AI into digital pathology could further refine risk assessment."

2) Implications and Utility

The authors suggest that the presented system could provide better patient stratification than current methods, such as gene expression profiling or clinical features. For example, they state:

"The discrepant results of different tests and their restrictive indications limit their use, while their cost is currently not covered by the health care system in many countries."

"Our system could support decision-making, replace molecular signatures or, at least, work as a pre-screening test."

However, the paper does not provide evidence for these claims:

- There is no comparative analysis showing improvement over existing molecular tests (e.g., Oncotype DX).
- The authors do not present cost-effectiveness data to support their claim, which is especially relevant in countries without genomic test coverage.
- There is no discussion on the methodology of pre-screening or threshold setting for endorsing genomic tests. One would need to define a threshold below or above which patients would be excluded/endorsed for the expansive genomic tests.

We acknowledge that this part of the discussion was not fully supported by the results presented. To provide a perspective on the potential pre-screening application of our test without overstating its clinical utility— which has yet to be established— we have substantially revised this section. We have softened our critique of genomic tools, refined our wording, and incorporated a new reference on digital pathology for pre-screening (Boehm, Kevin M., et al. 'Multimodal histopathologic models stratify hormone receptor-positive early breast cancer.' BioRxiv (2024): 2024-02). The revised paragraph now reads as follows:

"Multimodal histopathologic models have been investigated as a means to stratify hormone receptor-positive early breast cancer and could serve as pre-screening tools to identify patients who may or may not require more expensive genomic testing

(Boehm et al 2024). While our study does not provide a direct comparison with genomic assays such as Oncotype DX, our results suggest that a WSI-based approach could serve as a complementary tool in clinical decision-making. Head-to-head comparison of RlapsRisk BC and molecular signatures in a cohort similar to TRANSATAC would provide further insights into its potential clinical utility, including defining optimal thresholds for genomic test endorsement (Sestak et al. 2018). In addition, assessment of the impact of a decision-making strategy based on RlapsRisk BC would require a randomized clinical trial to eliminate potential biases that may arise from observational studies.”

- The authors do not discuss the limitations of AI models for clinical decision-making, including interpretability and the risks of domain shift.

Domain shift is indeed a significant concern in digital pathology, and we now explicitly address it in the Discussion as follows: “To use RlapsRisk BC in clinical practice, the calibration protocol requires 30 BC slides, which may pose challenges, particularly in smaller pathology laboratories. While calibration is a standard procedure for many medical devices, it could impair the clinical deployment of such AI tools. We have added the following paragraph in the discussion section

“Furthermore, variations in staining protocols, scanner settings, and other technical factors between healthcare centers could introduce domain shifts, potentially affecting the model’s reliability. Here prognosis results were robust with respect to the staining protocols (H&E and HES) since concordant prognostic results were found in two centers using two different staining protocols. However, the diversity of worldwide staining protocols is larger than the two protocols tested here, stressing the importance of analytical validation.”

Interpretability is already addressed in another paragraph and we now have considerably revised the paragraph about interpretability that now reads as follows

“The interpretability analysis of the pathological features associated with an increased or decreased risk of relapse, as predicted by RlapsRisk BC, supports the validity of our model. The model’s reliance on well-established histological prognostic factors—such as nuclear pleomorphism, tumor architecture, and mitotic activity—demonstrates its alignment with known pathology-driven risk assessments.”

- Without randomized treatment studies, a prognostic model has limited utility, as confounding factors may prevent actionable outcomes.

We add the following sentence to acknowledge the limitation of our study **“In addition, assessment of the impact of a decision-making strategy based on RlapsRisk BC would require a randomized clinical trial to eliminate potential biases that may arise from observational studies.’ to acknowledge this limitation.”**

- Finally, the authors refrain from publishing their data, model, or code, which further limits the implications of their work.

The scripts used to run the training and inference of the featured model are not public as they rely heavily on the intellectual property of Owkin, Inc. and on proprietary libraries. Those libraries include a tool to preprocess and tile raw whole-slide images; and a library of weakly supervised machine learning models. Nonetheless we provided the scripts used to generate the results presented in our manuscript.

3) Methodology

The methodology relies heavily on existing techniques:

- Feature extraction is based on SSL (43), and the downstream task uses the well-established MIL strategy (46) and smoothed c-index loss (47). These approaches are not novel.

Our study focuses on the extensive evaluation of RlapsRisk BC in a clinical setting rather than developing novel ML techniques. While we build on established methods like SSL (43), MIL (46), and smoothed c-index loss (47), our key contribution lies in demonstrating the real-world applicability of AI-driven prognostic modeling through rigorous validation.

- The authors do not clarify why they cannot publish their model or code if it is based on prior methods.

As already mentioned in a previous review and again here at the end of “2) Implications and Utility”, we are constrained from publishing the model or code due to intellectual property issues (Owkin, Inc. being a private company). We understand the importance of transparency and replicability in scientific research. While we are unable to release the code due to proprietary reasons, we are committed to upholding the highest standards of integrity and rigor in our methodologies. We recognize the significance of replicability in advancing the field and will continue to explore ways to contribute to this goal within the constraints of our business model. It's essential to address the concerns raised, and we remain dedicated to mitigating any issues identified to ensure the credibility and trustworthiness of our work.

- The SSL feature extractor was changed during the review process: "We thank the reviewers for this valuable feedback which led us to retrain new feature extractors using new state-of-the-art methods: IBOT ViT Base."

This raises the question: What exactly is RlapsRisk® BC? Is it an established method?

RlapsRisk BC is an evolving AI-based prognostic tool, and like many IVD SaMD (In Vitro Diagnostic Software as a Medical Device) solutions, it undergoes continuous improvement. The version presented in this study reflects the most up-to-date model, incorporating advancements such as IBOT ViT Base for SSL feature extraction.

A major concern is the lack of clarity about how the combined model using both image-based and clinico-pathological features was created and trained. Were these features combined during training? If it was evaluated together in the multivariate analysis on the test set, the analysis may not be sound.

We acknowledge that the presentation of the Combined score was not clear enough. We added a few sentences to precise how the Combined score was obtained, it now reads as follows in the *Model development and datasets* section of the Results: **“The coefficients of this Cox model, which provides the Combined Score, were also trained on the discovery dataset.”** We confirm that the Combined score is obtained by using only models fitted on the discovery dataset.

To avoid any misunderstanding, we also added a precision “RlapsRisk BC score and Prognosis” section of the Results on the multivariable analyses, where indeed the HRs are computed using the external validation cohort : **“RlapsRisk BC score was an independent prognostic factor of MFS (Table 1A and Supplementary Table 3A) when integrated into a multivariable Cox model including histological grade, age, lymph node invasion, tumor size, and Ki67 expression directly fitted on cohorts”.**

4) Soundness of Analysis and Reproducibility

The authors' analysis appears valid and provides evidence for their main claim that H&E/HES images add prognostic information over clinico-pathological variables using the proposed model. Specifically, regarding reviewer 2's comment on the 3-fold cross-validation, the newly proposed method of using 10,000 permutations to compute the p-value is valid. However, several issues remain:

- The fact that the code, models, and data are not published significantly limits the soundness and reproducibility of the study.
- The test on the CANTO dataset involves a mix of two populations (different slides and cohorts), rather than analyzing each cohort separately, which could introduce confounding factors.

As described in the supplementary material, separate analyses yield results consistent with those obtained from the entire cohort. These findings are presented in the supplementary section titled 'Results on the CANTO Cohort per Staining Method,' specifically in Tables 4A and 4B, which detail the multivariable analyses stratified by population.

- Details on the analysis are incomplete. For example:

There is insufficient information on the training process. The authors mention: "RlapsRisk BC scores are derived from the validation splits of the repeated cross-validation training procedure," but there is no detailed discussion of cross-validation, folds, or splits across the paper.

We acknowledge that the details on the training procedure were missing, it now reads as follows : **“The RlapsRisk BC risk score was optimized using the discovery cohort GrandTMA (HES WSIs) with a stratified fivefold cross-validation approach, repeated three times. Given the limited number of events, stratification was performed based on event occurrence to ensure a minimum number of events in each fold.”**

o The authors refer readers to another paper to understand the model and loss function. However, these details are critical for understanding and replicating the study, and their absence in the manuscript is a notable shortcoming.

A more detailed description of the functioning DeepMIL has been added to the section called “Risk Prediction using Multiple Instance learning” of the methods, it now reads as follows : **“DeepMIL is a deep multiple instance learning (MIL) framework designed for weakly supervised classification. It aggregates instance-level (the WSI’s tiles in our case) features using an attention-based pooling mechanism, which assigns learned importance weights to each instance within a bag. This allows the model to focus on the most relevant instances for prediction. The weighted instance representations are then combined into a single bag-level feature vector, which is processed by a fully connected layer to generate the final classification output. This approach enhances interpretability while effectively handling variability within each bag.”**

We also added a description of the functioning of the loss function in the same paragraph, it now reads as follows: **“We used a smooth differentiable loss function based on a sigmoid approximation of the standard C-index indicator function. This smoothing introduces a parameter σ that controls the transition sharpness, enabling gradient-based optimization. The loss function is integrated into a gradient boosting framework, iteratively minimizing the smoothed empirical risk to improve model discrimination. This approach ensures that the learned biomarker combinations are directly aligned with survival prediction performance.”**

5) Writing Quality

Although this is not a major concern, I found the paper to be not fluent, with missing details, difficult to read, and occasionally containing typos. Some examples include:

- Unclear sentence structure:

o "This higher risk of the patients with chemotherapy being captured by the clinical variables in our multivariable analyses, we excluded the variable representing the adjuvant chemotherapy treatment regimen from all prognostic models to avoid redundancy of information that could lead to multicollinearity issues."

This sentence structure is convoluted and unclear.

It has been modified and now reads as follows **“We excluded chemotherapy treatment from the clinical score because it was prescribed to higher-risk patients based on clinical best practices, introducing prescription bias. Since this risk was already captured through clinical variables, including chemotherapy would have led to redundancy and multicollinearity in the prognostic models.”**

o "We considered this score as our baseline reference that we compared to a score based on a Cox Model adjusted for the clinical factors and RlapsRisk BC score to assess the additional prognostic value of the RlapsRisk BC score to the standard clinical factors."

It has been modified and now reads as follows: **“We used this score as the baseline reference and compared it to a score derived from a Cox model adjusted for clinical factors and the RlapsRisk BC score (referred to as the Combined Score).”**

This sentence is unnecessarily long and difficult to follow.

- Misleading terminology:

- o If the data consists of whole slide images, why is the cohort referred to as "GrandTMA"? The name is misleading and could cause confusion for readers expecting tissue microarray (TMA) data.

The training cohort chosen for this study originates from an institutional project at Gustave Roussy with the aim of creating a large tissue microarray that could allow the discovery and/or validation of biomarkers in a substantial subset of breast cancer samples. We found that this population and the available material (clinical data, histopathological glass slides stored in the pathology department) were perfectly suited to our purpose. Therefore, the name refers to the primary objective of this database and we do not think that we could rename this collection just because we are now using it for a different application than originally designed.

There are many similar instances where clarity could be improved, and addressing these issues would significantly enhance the readability and accessibility of the paper.

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Reviewer #6 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

We appreciate the feedback provided by you and the reviewers and the opportunity to clarify key aspects of our study.

Response to Referees #2

Dear Dr Garberis,

Thank you again for submitting your revised manuscript "Deep Learning Assessment of Metastatic Relapse Risk from Digitized Breast Cancer Histological Slides." to Nature Communications. We have now received reports from the reviewers who evaluated the original version. On the basis of their comments (copied below), we have decided to invite an additional revision of your work.

You will see that, while the reviewers find that your revisions improved the manuscript, some important points remain to be addressed. In particular, we would expect your revision to address Reviewer #5's remaining concerns regarding methods details throughout. Please revise your manuscript, addressing all the remaining issues raised by the reviewers.

We appreciate the valuable feedback from the reviewers and the opportunity to clarify key aspects of our study. Below, we address the specific concerns raised, including:

- Selection of model hyperparameters and performance comparisons
- Memory management
- Overall pipeline applied
- Additional details for clarity and reproducibility

REVIEWER COMMENTS

Reviewer #4 (Remarks to the Author):

The authors addressed all my comments.

We appreciate the reviewer's insightful comments and suggestions.

Reviewer #5 (Remarks to the Author):

I appreciate the authors' detailed responses and the clarifications in the Discussion section. However, I still have concerns about the methodological details, which are critical for reproducibility—especially since the authors are unable to publish their code. Below are specific points that need further elaboration:

In a smoothed c-index approach, the computational complexity is $O(N^2)$ in the number of slides N . When each slide has many tiles, memory usage can become prohibitive. There is also the practical requirement of ensuring that each minibatch contains enough events for stable gradient updates. If events occur only every 20–30 patients, what was the average number of events per minibatch in practice and what was the minibatch size?

Clarifying how the minibatch size and tile sampling per slide were chosen and how memory constraints were mitigated would help readers reproduce the method.

We thank the reviewer for highlighting this important aspect. To ensure stable gradient updates, we used a minibatch size of 128 patients *as specified in the Methods section “Risk prediction using Multiple Instance Learning”*, randomly sampled for each iteration. **Given the 7.69% relapse rate in our training cohort, this resulted in an average of 9.84 events per minibatch, providing sufficient supervision for model optimization. While minibatches without observed events were possible, their occurrence was rare (~0.0036% probability). Although such batches do not directly contribute to differentiating survival outcomes, their impact was mitigated by the smoothed C-index loss, which allows gradient updates even when minibatches contain few or no events.**

Regarding tile sampling per slide during training, as detailed in the same Methods section, we randomly selected a subset of 8,000 tiles per whole-slide image (WSI), whereas all tiles from a slide were processed during external validation. This approach allowed to accelerate model training while maintaining a representative overview of the entire slide, as adjacent tiles often contain similar biological information. Additionally, this strategy allowed us to fit all relevant features into memory without compromising the model's ability to capture meaningful patterns.

From a performance perspective, a batch size of 128 provided the best results in cross-validation:

Batch Size	Cross-validation AUC
128	0.750 ± 0.084
64	0.734 ± 0.086
32	0.748 ± 0.081
16	0.742 ± 0.087

To assess the impact of the number of tiles per patient's WSI used during training, we conducted a benchmark analysis. Four random samplings were performed to select tile subsets for each patient, and performance was averaged across these subsets:

Number of tiles per patient	Cross-validation AUC
8000	0.750 ± 0.084
3000	0.742 ± 0.097
1000	0.735 ± 0.084

How exactly was the end-to-end pipeline set up?

We added the following precisions in the manuscript : **“The models were trained for 15 epochs with a batch size of 128. Each epoch required approximately 26 seconds to complete, resulting in a total training time of 6 minutes and 30 seconds per fold. The entire training of the models with five times repeated 3-fold cross-validation process with multiple random weight initializations took approximately 1 hour 38 minutes. Training was conducted on a Tesla T4 Nvidia GPU using the PyTorch framework. The**

CPU memory usage for loading all features of the training data was 45GB, with an additional 13GB of GPU memory utilized during training.”

Were there any techniques such as pairwise sub-sampling for the c-index, or any special strategies to handle large risk sets?

No specific upsampling of events in batch training was used as precised in the upper response. The incidence of events in each batch was proportional to the incidence of the whole cohort (7.69%).

The paper should specify any memory-optimization tricks.

We added the following description in the manuscript : **“Training was conducted on a Tesla T4 Nvidia GPU using the PyTorch framework. The CPU memory usage for loading all features of the training data was 45GB, with an additional 13GB of GPU memory utilized during training.”**

By using only the features of 8,000 tiles per patient slide, we were able to fit all training data into memory, thereby accelerating the training process. No additional methods were employed for memory optimization (see upper response for benchmark).

The authors mention using a smoothed c-index loss, but it would be helpful to understand the reasons for preferring this over a Cox partial-likelihood loss.

We opted for the smoothed c-index loss during training because it has been shown to enhance model performance, particularly when combined with stability selection. As demonstrated in the study published in BMC Bioinformatics (<https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-016-1149-8#Sec7>), the C-index boosting method can yield models with superior discriminatory power. This suggests that directly optimizing the C-index can lead to better prognostic predictions compared to relying solely on Cox partial-likelihood loss. In our own comparisons, the results were nearly equivalent, with the smoothed c-index loss achieving a score of 0.750 ± 0.084 versus 0.745 ± 0.092 for the Cox partial-likelihood loss in cross-validation on our dataset, and we thus preferred to select the smoothed c-index loss for the training of the final models.

Overall, the study's topic is important, and the authors' work could be valuable if these clarifications are provided. Without these methodological details, reproducing and extending the findings would be challenging.

To enhance clarity and facilitate the reproducibility of our methodology, we have added a section titled “Cross-validation Experiments and Model Hyperparameters” in the supplementary methods, consolidating all relevant information. Additionally, several details have been incorporated into the main manuscript (highlighted in blue) to further support this objective.

We sincerely thank the reviewer for their valuable feedback, which has helped improve both the quality of the paper and its clarity for the reader.

Reviewer #5 (Remarks on code availability):

See above

Reviewer #6 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #6 (Remarks on code availability):

The methodological code was not available

[We thank the reviewer for its valuable comments and suggestions.](#)